

Product Requirements Document (PRD)

E-Commerce Platform

Document Version: 1.0

Product Owner: Product Manager

Status: Draft

Technology Context: Backend foundation implemented using Java + Spring Boot

Existing Domain Models: Product, Order, User

1. Product Vision

Build a scalable e-commerce platform that enables customers to discover products, manage their accounts, place orders, and track purchases while providing administrators with tools to manage inventory, products, and customer orders.

The platform should provide a secure, reliable, and high-performance shopping experience similar to modern commerce platforms.

2. Business Objectives

Primary Goals

1. Enable customers to:
 - Browse available products
 - Search and filter products
 - Manage shopping carts
 - Place orders
 - Track order status
 - View order history
2. Enable business operators to:
 - Manage product catalog
 - Manage inventory
 - Process customer orders
 - Analyze sales activity
3. Build a foundation that can scale:

- From thousands to millions of users
 - From hundreds to millions of products
 - From simple CRUD to event-driven commerce workflows
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3. Current Implementation Status

The following core entities have already been implemented using Spring Boot and Java.

Completed Domain Models

User Entity

Represents customers and platform users.

Current responsibilities:

- User registration
- User authentication data
- User profile information
- Customer identity management

Product Entity

Represents items available for purchase.

Current responsibilities:

- Product information
- Product pricing
- Product availability

Order Entity

Represents customer purchases.

Current responsibilities:

- Order creation
 - Order lifecycle management
 - Association between users and purchased products
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4. Scope

In Scope

Customer Features

- User registration/login
- Product browsing
- Product search
- Product details
- Cart management
- Checkout
- Order placement
- Order tracking
- Order history

Admin Features

- Product management
- Inventory management
- Order processing
- User management
- Sales reporting

Platform Features

- Authentication
- Authorization
- Payment integration
- Notification system
- Audit logging

Out of Scope (Phase 1)

- Marketplace sellers
- Recommendation engine
- Loyalty points

- Subscription products
 - International shipping
 - Multi-currency support
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5. User Personas

Persona 1: Customer

Name: Online Shopper

Goals

- Find products quickly
- Purchase securely
- Track deliveries

Pain Points

- Difficult navigation
 - Slow checkout
 - Lack of order visibility
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Persona 2: Administrator

Name: Store Manager

Goals

- Maintain product catalog
- Process orders
- Monitor sales

Pain Points

- Manual inventory updates
 - Lack of reporting
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6. Functional Requirements

6.1 User Management

Feature: User Registration

Description

Customers should be able to create an account.

Requirements

ID	Requirement
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USER-001	System shall allow users to register using email and password
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USER-002	Email must be unique
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USER-003	Password must be securely encrypted
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USER-004	User profile information shall be stored
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Acceptance Criteria

Given a new customer

When they submit valid registration details

Then an account should be created successfully.

Feature: User Authentication

Requirements

ID	Requirement
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USER-005	User shall login using credentials
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USER-006	System shall generate authentication token
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USER-007	Unauthorized access shall be blocked
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Recommended implementation:

- Spring Security
- JWT authentication
- Refresh tokens

6.2 Product Catalog

Feature: Product Listing

Customers can browse products.

Requirements

ID	Requirement
PROD-001	Display available products
PROD-002	Support pagination
PROD-003	Display price and availability
PROD-004	Allow filtering by category
PROD-005	Allow sorting by price/rating

Feature: Product Search

Requirements:

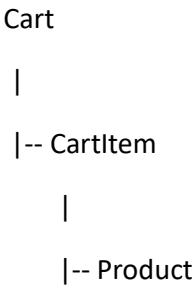
ID	Requirement
PROD-006	Search products by name
PROD-007	Search should support partial matching
PROD-008	Search response should complete within 500ms

Future enhancement:

- Elasticsearch integration
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6.3 Shopping Cart

New Entity Required



Requirements

ID	Requirement
CART-001	User can add products to cart
CART-002	User can update quantity
CART-003	User can remove products
CART-004	Cart should persist across sessions

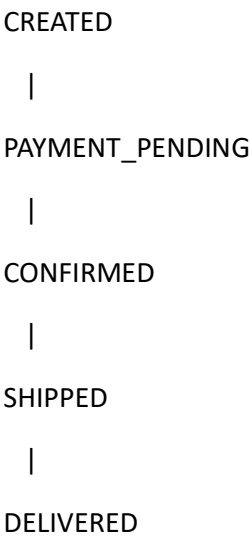
Cart Business Rules

- 1. Product quantity cannot exceed inventory.
 - 2. Out-of-stock products cannot be checked out.
 - 3. Cart price should always reflect latest product price.
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6.4 Order Management

Existing Order entity will be enhanced.

Order Lifecycle



Alternative paths:

CREATED

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CANCELLED

Requirements

ID	Requirement
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ORDER-001	User can create order from cart
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ORDER-002	Order should contain purchased products
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ORDER-003	Order price should be immutable after creation
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ORDER-004	User can view order history
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ORDER-005	User can cancel eligible orders
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Order Data Requirements

Order should store:

Order

id

userId

orderDate

status

totalAmount

paymentStatus

OrderItem

id

orderId

productId

productNameSnapshot

priceAtPurchase

quantity

6.5 Payment Management

New Entity Required

Payment

Requirements

ID	Requirement
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PAY-001	Support online payment
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PAY-002	Store payment transaction details
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PAY-003	Handle payment failures
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PAY-004	Support payment retry
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Possible integrations:

- Razorpay
 - Stripe
 - PayPal
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6.6 Inventory Management

New Entity Required

Inventory

Requirements

ID	Requirement
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INV-001	Track available stock
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INV-002	Reduce inventory after successful order
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INV-003	Prevent overselling
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INV-004	Notify low stock
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6.7 Notification System

Events

System should send notifications for:

Event	Notification
Registration	Welcome email
Order created	Order confirmation
Payment success	Receipt
Shipment	Tracking update
Delivery	Delivery confirmation

7. Non Functional Requirements

Performance

Requirement	Target
API response time	<500ms
Product search	<300ms
Checkout	<2 seconds
Availability	99.9%

Scalability

System should support:

Initial:

- 100K users
- 1M products
- 10K orders/day

Future:

- 10M users
 - 100M products
 - 1M orders/day
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Security

Requirements:

- HTTPS only
- Password hashing using BCrypt
- JWT authentication
- Role-based authorization

Roles:

CUSTOMER

ADMIN

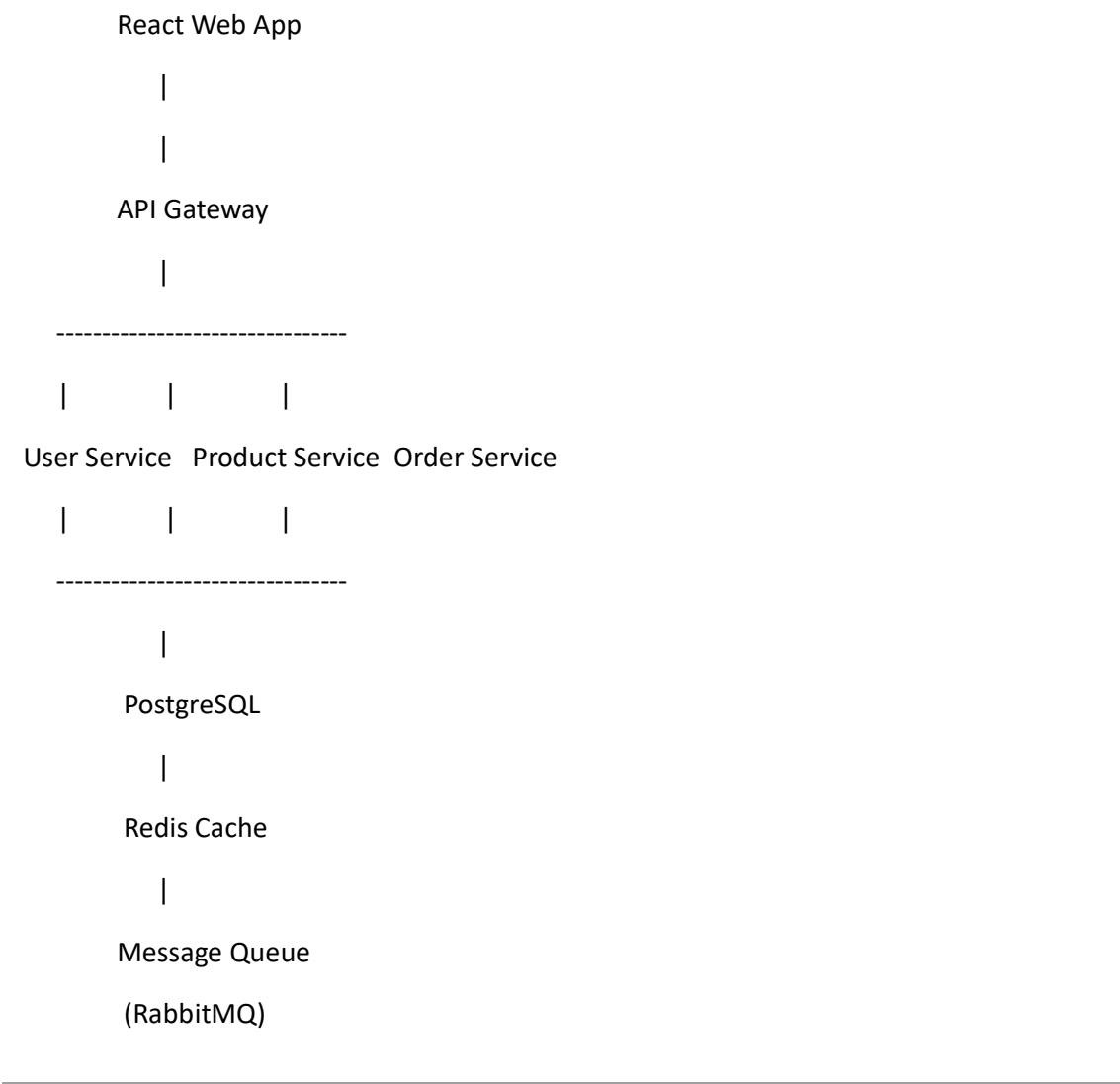
INVENTORY_MANAGER

Reliability

Requirements:

- Database transactions for order creation
 - Retry failed payments
 - Idempotent APIs
 - Audit logs
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8. Proposed System Architecture



9. Database Model Evolution

Current:

User

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Order

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Product

Future:

User

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+---- Cart

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+---- Order

|

+---- OrderItem

|

Product

Product

|

+---- Inventory

Order

|

+---- Payment

Order

|

+---- Shipment

10. REST API Requirements

User APIs

POST /api/users/register

POST /api/auth/login

GET /api/users/{id}

PUT /api/users/{id}

Product APIs

GET /api/products

GET /api/products/{id}

POST /api/products

PUT /api/products/{id}

DELETE /api/products/{id}

Cart APIs

GET /api/cart

POST /api/cart/items

PUT /api/cart/items/{id}

DELETE /api/cart/items/{id}

Order APIs

POST /api/orders

GET /api/orders/{id}

GET /api/users/{id}/orders

PUT /api/orders/{id}/cancel

11. Product Roadmap

Phase 1: MVP

Duration: 8-12 weeks

Deliver:

- User management
 - Product catalog
 - Cart
 - Checkout
 - Orders
 - Basic payment
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Phase 2: Growth

Deliver:

- Search engine
 - Recommendations
 - Reviews
 - Coupons
 - Analytics
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Phase 3: Enterprise Scale

Deliver:

- Microservices migration
- Event-driven architecture
- Multi-region deployment
- AI personalization

12. Success Metrics

Customer Metrics

- Conversion rate
- Cart abandonment rate
- Repeat purchase rate
- Average order value

Platform Metrics

- API latency
- Error rate
- Order success percentage
- System uptime

13. Open Product Decisions

Decision	Owner
Payment provider	Business
Shipping integration	Operations
Product categories model	Product
Inventory reservation strategy	Engineering
Monolith vs microservices migration Architecture	

Final Product Requirement Summary

The platform should evolve from the existing Spring Boot implementation of **User, Product, and Order** into a complete commerce ecosystem by introducing:

- Cart management
- Inventory control
- Payment processing
- Notifications
- Search
- Authentication
- Admin operations

The architecture should prioritize correctness in transactions first, then optimize for scale using caching, messaging, and service decomposition as traffic grows.